



Faculty of Engineering & Technology
Department of Electrical and Computer Engineering
ENCS5341 – Machine Learning & Data Sceince
Report
Assignment #3

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I. Introduction

1.1 Task Description

The chosen Task is to predicts weather conditions from outdoor images by classifying them into predefined weather categories. The task is challenging due to real-world variability in lighting, geographic context, and environmental conditions, as well as significant class imbalance. Exploratory data analysis highlighted additional challenges, including overlapping visual cues between weather classes and uneven geographic and category distributions.

1.2 Models Explored

Multiple machine learning models were explored to address the image-based weather prediction task, ranging from simple similarity-based baselines to models leveraging deep visual representations. As a baseline, a **k-nearest neighbor (kNN)** classifier was evaluated using **Euclidean distance** in the CNN feature space. Two configurations were considered (**k = 1** and **k = 3**) to provide reference performance without explicit model training.

A **pretrained convolutional neural network (CNN)** was then used to extract fixed-length feature vectors from images. These feature vectors capture high-level visual patterns such as textures and scene structure while significantly reducing the dimensionality of the raw image data.

Finally, the extracted CNN features were used to train a **Support Vector Machine (SVM)** classifier. The SVM was selected for its effectiveness in high-dimensional spaces, and its regularization parameter was tuned to evaluate its impact on classification performance.

1.3 Evaluation Metrics

Model performance was evaluated using multiple metrics together provide a balanced and informative evaluation framework, provide a comprehensive assessment.

- **Accuracy** was used to measure overall classification correctness.

- **Precision, recall, and F1-score** were reported for each class to evaluate performance on both frequent and rare weather categories.
- **Macro-averaged metrics** were emphasized to ensure that minority classes contribute equally to the evaluation.
- A **confusion matrix** was used to analyze common misclassifications and better understand model behavior across different weather conditions.

II. Exploratory Data Analysis

2.1 Dataset Structure and Descriptive Statistics

The dataset initially contained 713 samples with 9 attributes. During preprocessing and image–label validation, 688 images were successfully downloaded automatically, while 25 downloads failed due to security restrictions. These 25 images were then downloaded manually to complete the dataset. Most features are categorical with limited missing values.

First 5 rows of the dataset:

	StudentID	Image URL	Description	Country	Weather	Time of Day	Season	Activity	Mood/Emotion
0	1210241	https://littleholidays.net/wp-content/uploads/...	A scenic alpine village surrounded by green va...	Switzerland	Sunny	Afternoon	Summer	Sightseeing	Happiness
1	1210241	https://cdn.praguecitytourism.city/2024/03/131...	Historic cobblestone street with old buildings...	Czechia	Cloudy	Afternoon	Not Clear	Sightseeing	Nostalgia
2	1210241	https://i.pinimg.com/736x/8b/b7/21/8bb72102513...	Historic stone streets and old-town architectu...	United Kingdom	Snowy	Evening	Winter	Wandering	Romance
3	1210241	https://www.tripsavvy.com/thumb/C55SxvFfoqLpb9l...	Snow-covered city street of New York, with bul...	United States	Snowy	Evening	Winter	Wandering	Nostalgia
4	1210241	https://www.attica21hotels.com/wp-content/uplo...	Aerial view of Barcelonas rooftops and archite...	Spain	Sunny	Afternoon	Summer	Wandering	Curiosity

Figure 1: First five rows of the dataset.

2.2 Image Description Analysis (Input feature)

The W/H (aspect ratio) is the ratio of an image’s width to its height, describing its shape (e.g., 16:9, 4:3, 1:1) regardless of size. We plotted aspect ratios to show that our dataset contains images with different dimensions and shapes, which impacts preprocessing. We also displayed the first 10 images under Image Description Analysis to visually confirm this variation and check overall image quality. Since most vision models require a fixed input size and efficient batching needs identical dimensions, we apply resizing to standardize all images. To avoid stretching or squeezing objects, we prefer resizing methods that preserve the original aspect ratio.

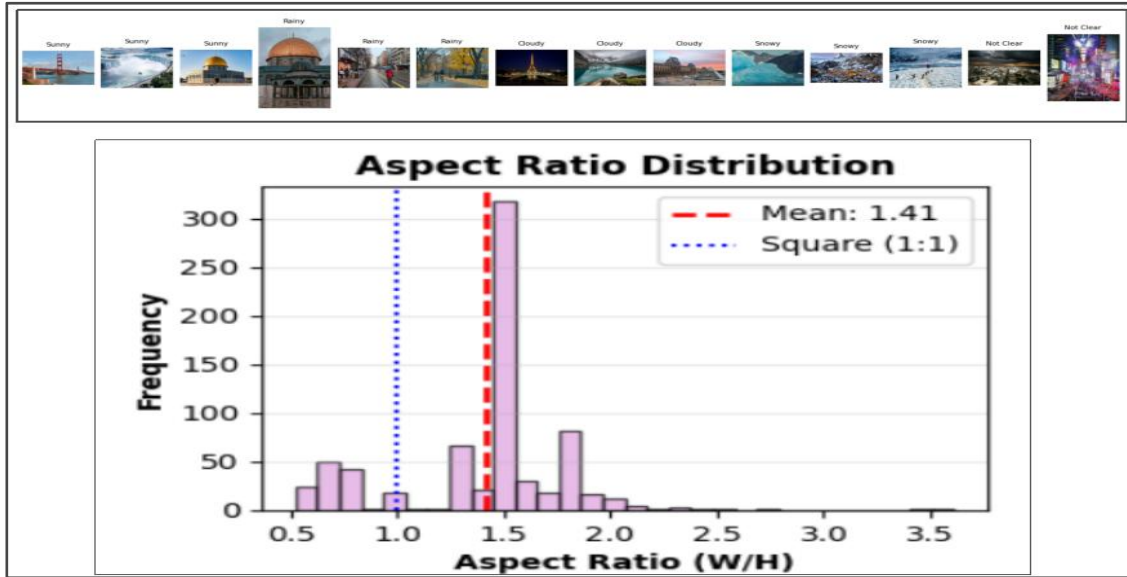


Figure 2: First 10 dataset image samples and the aspect ratio (W/H) distribution.

2.3 Weather Description Analysis (Target feature)

This section analyzes the target variable Weather to understand the class distribution before modeling. The pie chart and bar chart show a clear class imbalance, where Sunny is the dominant category (more than half of the samples), followed by Cloudy and Not Clear, while Snowy and especially Rainy appear much less frequently. This imbalance is important because it can bias the classifier toward predicting the majority class, so it should be addressed during training using techniques such as class weighting, balanced sampling, or careful evaluation metrics (e.g., macro F1-score) to ensure fair performance across all weather classes.

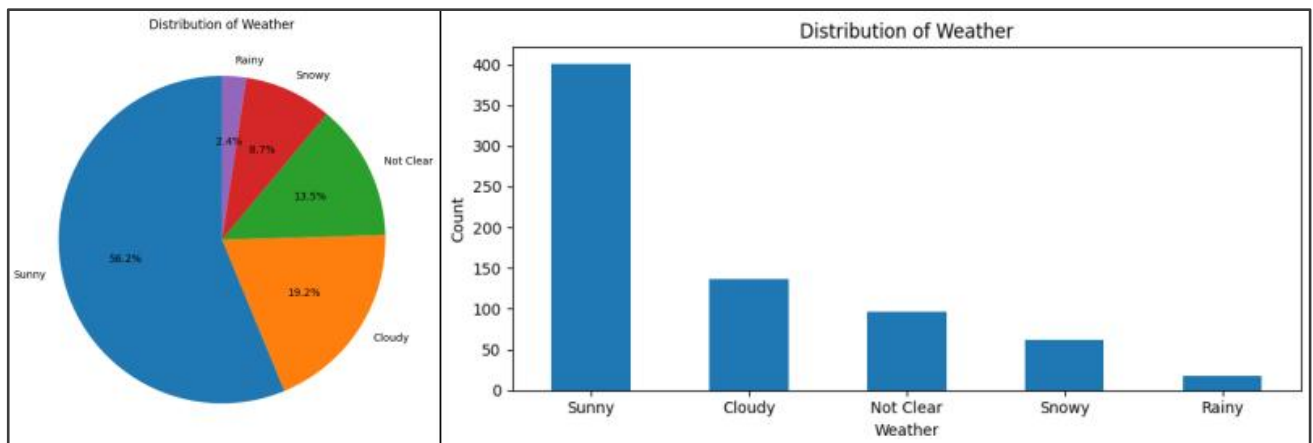


Figure 3: Distribution of the Weather classes (pie and bar charts).

2.4 General Trends in data (other features)

The distributions of the contextual features help explain the target Weather patterns, but they do not map one-to-one. For example, Summer is the most frequent season and this aligns with Sunny being the dominant weather class, suggesting that many images were captured in conditions typically associated with clear skies. Similarly, the higher counts for Afternoon and Evening are consistent with more outdoor, well-lit scenes where sunny or partly cloudy conditions are common.

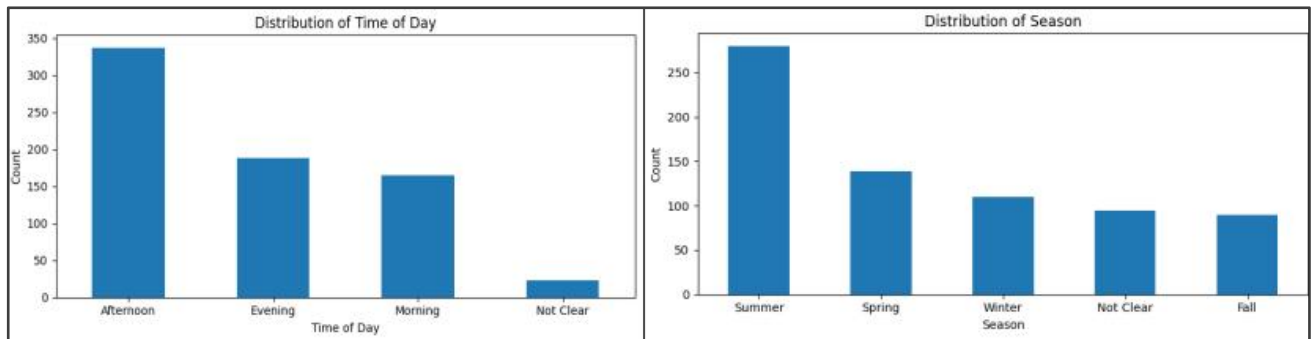


Figure 4: Distributions of key contextual features (Time of Day and Season).

However, the proportions are not identical because weather is also influenced by geography and local climate: an image can be labeled Snowy even during “Summer” if it was taken in high-altitude mountainous regions, and “Not Clear” or “Cloudy” can appear in any season or time of day due to fog, storms, or regional conditions. This shows that while season and time-of-day provide useful context, the dataset reflects real-world variability, so Weather must be learned from both visual cues and contextual metadata rather than assuming a fixed seasonal rule.

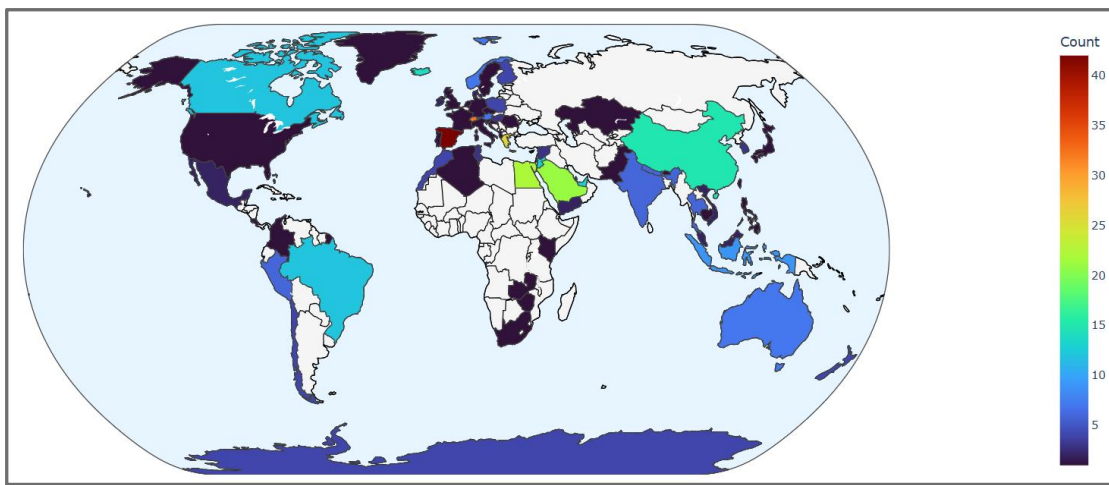


Figure 5: Geographic distribution of images by country (world map).

2.5 Correlations

The Cramér's V heatmap shows the strength of association between categorical variables (higher values = stronger relationship). Most pairs have **weak to moderate** associations, meaning the features are not redundant and can add complementary information. **Weather** is most strongly linked to **Season**, and also shows a noticeable association with **Country**, reflecting seasonal and geographic climate effects. In contrast, **Mood/Emotion** has a weaker connection to Weather, suggesting it is less directly related to meteorological conditions.

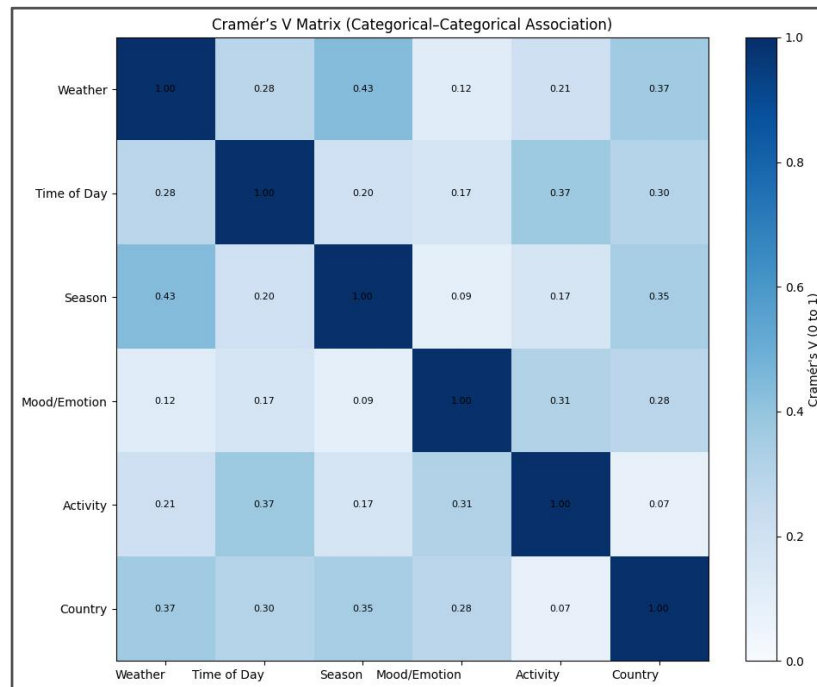


Figure 6: Cramér's V correlation matrix.

Weather $\times$ Season Distribution (%) heatmap shows how weather classes are distributed within each season. **Summer** is largely **Sunny**, while **Winter** contains a much higher share of **Snowy**, which matches real-world expectations. **Spring** and **Fall** appear more mixed, with more **Cloudy/Not Clear** cases. The presence of non-typical combinations (e.g., cloudy in summer) indicates realistic variability driven by geography and local conditions, so Season is helpful context but not a strict rule.

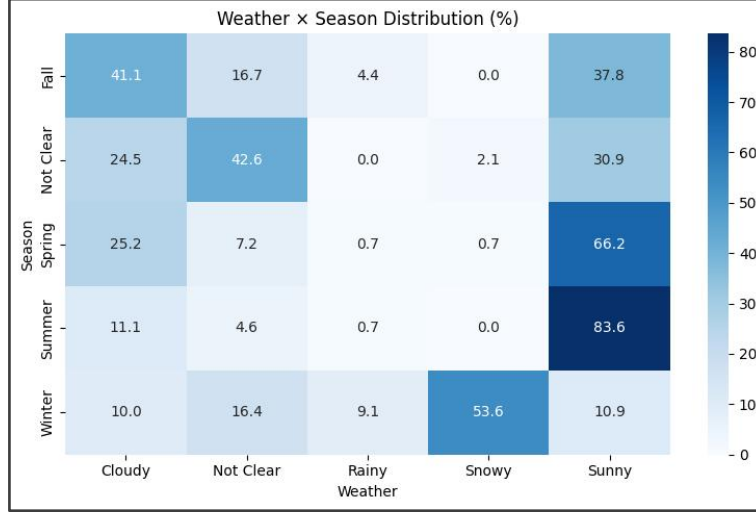


Figure 7: Weather $\times$ Season distribution (%).

III. Experiments and Results

1. Experimental Setup

To evaluate our ability to predict weather conditions from images, we trained and tested multiple models using the same dataset split to ensure a fair comparison. The dataset contains 717 labeled images across five weather categories (Sunny, Cloudy, Snowy, Rainy, Not Clear). Images were downloaded successfully from URLs and labels were standardized into consistent classes. The data was then split into 573 training images and 144 test images.

2. Baseline Model: K-Nearest Neighbors (KNN)

2.1 Feature Extraction

Since KNN cannot work directly on raw images, each image was converted into a numerical feature vector using hand-crafted visual features. These features were designed to capture important weather-related cues, including:

- **Color features:** RGB mean/std, brightness, and channel ratios
- **Color histograms:** intensity distribution per channel
- **Texture features:** gradients and entropy to represent roughness and cloud patterns

- **Sky-region features:** statistics from the upper image region (where sky usually appears), including “blue-ness” ratios

These features were standardized using scaling before training.

2.2 Baseline Results

We experimented with two values of k : $k = 1$ and $k = 3$, where $k = 3$ is selected as the main baseline due to stronger performance.

```

KNN (k=1):
Test Accuracy: 61.11%
CV Score: 62.83% (+/- 2.18%)

Classification Report:

```

	precision	recall	f1-score	support
Cloudy	0.30	0.39	0.34	28
Not Clear	0.00	0.00	0.00	0
Rainy	0.50	0.33	0.40	3
Snowy	0.33	0.27	0.30	11
Sunny	0.78	0.72	0.75	102
accuracy			0.61	144
macro avg	0.38	0.34	0.36	144
weighted avg	0.65	0.61	0.63	144

Figure 8: KNN ($k=1$) baseline performance with classification report and accuracy.

```

KNN (k=3):
Test Accuracy: 70.14%
CV Score: 66.14% (+/- 1.56%)

Classification Report:

```

	precision	recall	f1-score	support
Cloudy	0.43	0.43	0.43	28
Not Clear	0.00	0.00	0.00	0
Rainy	1.00	0.33	0.50	3
Snowy	0.29	0.18	0.22	11
Sunny	0.81	0.84	0.83	102
accuracy			0.70	144
macro avg	0.51	0.36	0.40	144
weighted avg	0.70	0.70	0.70	144

Figure 9: KNN ($k=3$) baseline performance with classification report and accuracy.

The improvement from $k = 1$ to $k = 3$ indicates that using more neighbors helps reduce sensitivity to noisy images and mislabeled samples.

3. Proposed Model 1: Support Vector Machine (SVM)

3.1 Motivation

SVM was selected because it often performs well on small-to-medium datasets and is effective in separating classes in high-dimensional feature spaces. Since our hand-crafted feature vector already provides meaningful information, SVM is expected to outperform KNN by learning more structured decision boundaries.

3.2 Hyperparameter Tuning

The main tuned hyperparameter was the regularization parameter C , which controls the tradeoff between maximizing margin and minimizing classification error. We tested five values:

$$C \in \{0.1, 1, 10, 50, 100\}$$

For each value, both cross-validation accuracy and test accuracy were computed.

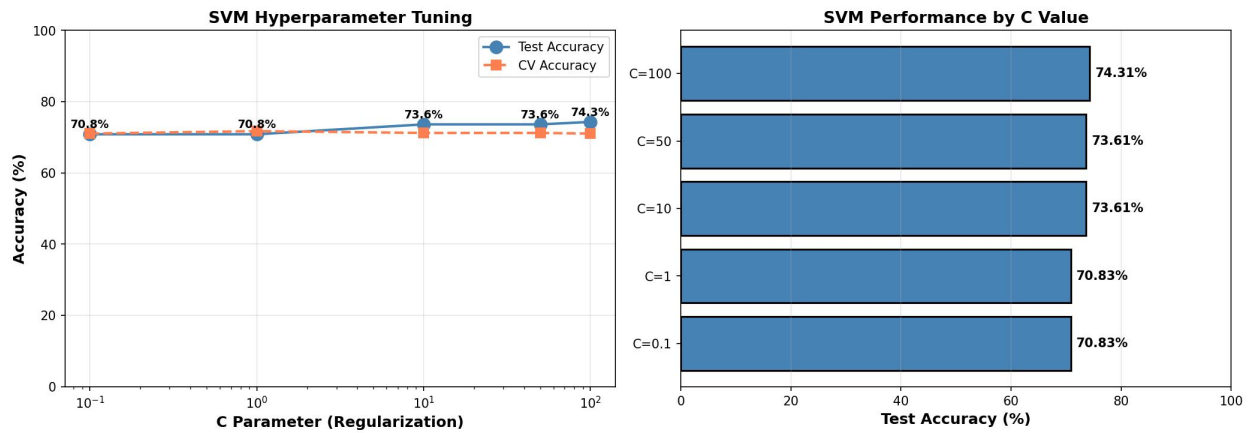


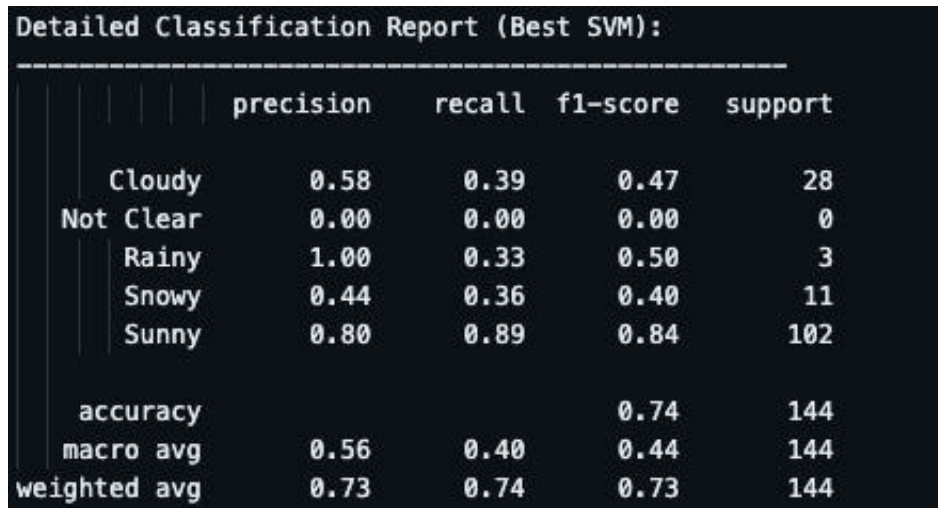
Figure 10: SVM hyperparameter tuning.

3.3 SVM Results

The SVM performance improved gradually as C increased. The best performance was achieved at:

- **Best C = 100**
- **Best SVM test accuracy = 74.31%**

This shows that allowing the model to fit the data more tightly (higher C) improved generalization for our extracted weather features.



```
Detailed Classification Report (Best SVM):
-----
              precision    recall  f1-score   support

   Cloudy         0.58        0.39        0.47         28
  Not Clear         0.00        0.00        0.00          0
     Rainy         1.00        0.33        0.50          3
     Snowy         0.44        0.36        0.40         11
     Sunny         0.80        0.89        0.84        102

 accuracy                   0.74         144
 macro avg         0.56        0.40        0.44         144
weighted avg         0.73        0.74        0.73         144
```

Figure 11: Classification report of the best SVM model (C=100).

4. Proposed Model 2: CNN using Transfer Learning (ResNet50)

4.1 Motivation

While KNN and SVM rely on manual feature design, deep learning models can automatically learn discriminative features from raw pixels. However, because our dataset is relatively small, training a CNN from scratch would likely cause overfitting. Therefore, we used transfer learning with a pretrained ResNet50 model (trained on ImageNet).

4.2 Architecture

The CNN model consisted of:

- ResNet50 base (frozen pretrained backbone)
- Custom classifier head:

$GAP \rightarrow Dense(512) \rightarrow Dense(256) \rightarrow Output(5)$

Total parameters: **24,769,413**

The model was trained with early stopping, and the best weights were restored automatically after training stopped at 11 epochs.

4.3 CNN Results

The CNN achieved:

- Test accuracy = 74.31%

This matches the best SVM performance, showing that pretrained CNN features can perform strongly even on limited weather image data.

Detailed Classification Report:					
		precision	recall	f1-score	support
	Cloudy	0.50	0.29	0.36	28
	Rainy	0.25	0.33	0.29	3
	Snowy	0.67	0.36	0.47	11
	Sunny	0.80	0.92	0.85	102
	accuracy			0.74	144
	macro avg	0.55	0.48	0.49	144
	weighted avg	0.72	0.74	0.72	144

Figure 12: : Classification report of the CNN model.

5. Comparison of All Models

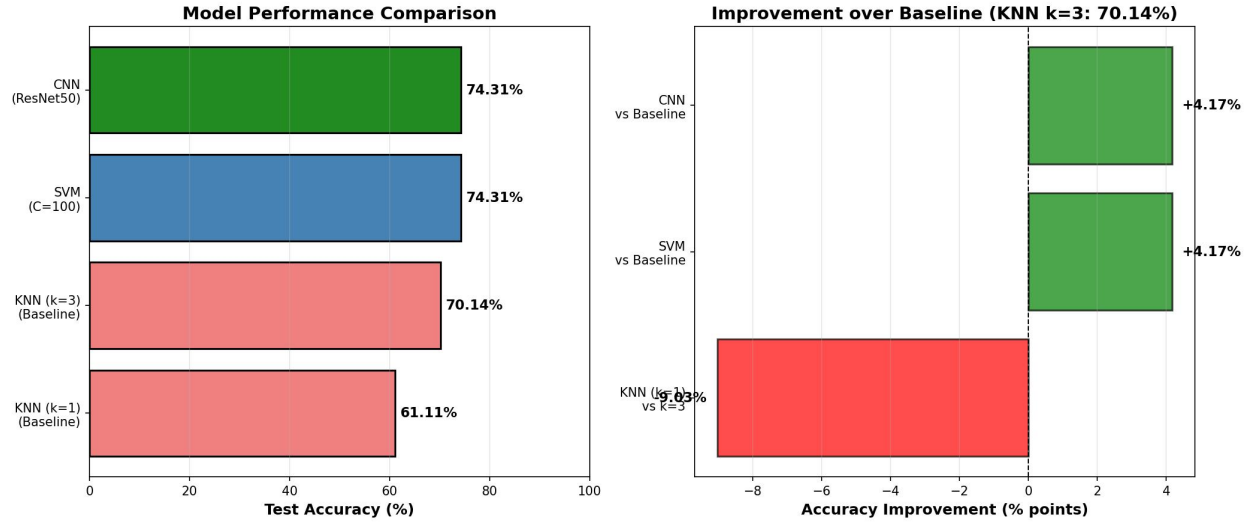


Figure 13: Model Performance Comparison & Improvement over Baseline

IV. Analysis

1. Overall Error Statistics

The CNN was evaluated on 144 test images, producing:

- Correct predictions: 107 (74.31%)
- Incorrect predictions: 37 (25.69%)
- Overall error rate: 25.69%

These values confirm that while the model performs reasonably well overall, a non-trivial portion of test samples are still misclassified, making it important to analyze the error distribution across weather categories.

2. Per-Class Error Analysis (Which labels are difficult?)

To detect whether errors are concentrated in specific categories, we computed per-class accuracy and error counts:

- **Sunny:** 92.16% accuracy (8 errors out of 102)
- **Cloudy:** 28.57% accuracy (20 errors out of 28)

- **Snowy:** 36.36% accuracy (7 errors out of 11)
- **Rainy:** 33.33% accuracy (2 errors out of 3)

The most difficult class was Cloudy with an error rate of 71.43%, while the easiest class was Sunny with an error rate of 7.84%. This indicates that the model is strongly biased toward recognizing Sunny conditions and struggles significantly on overcast and low-visibility categories.

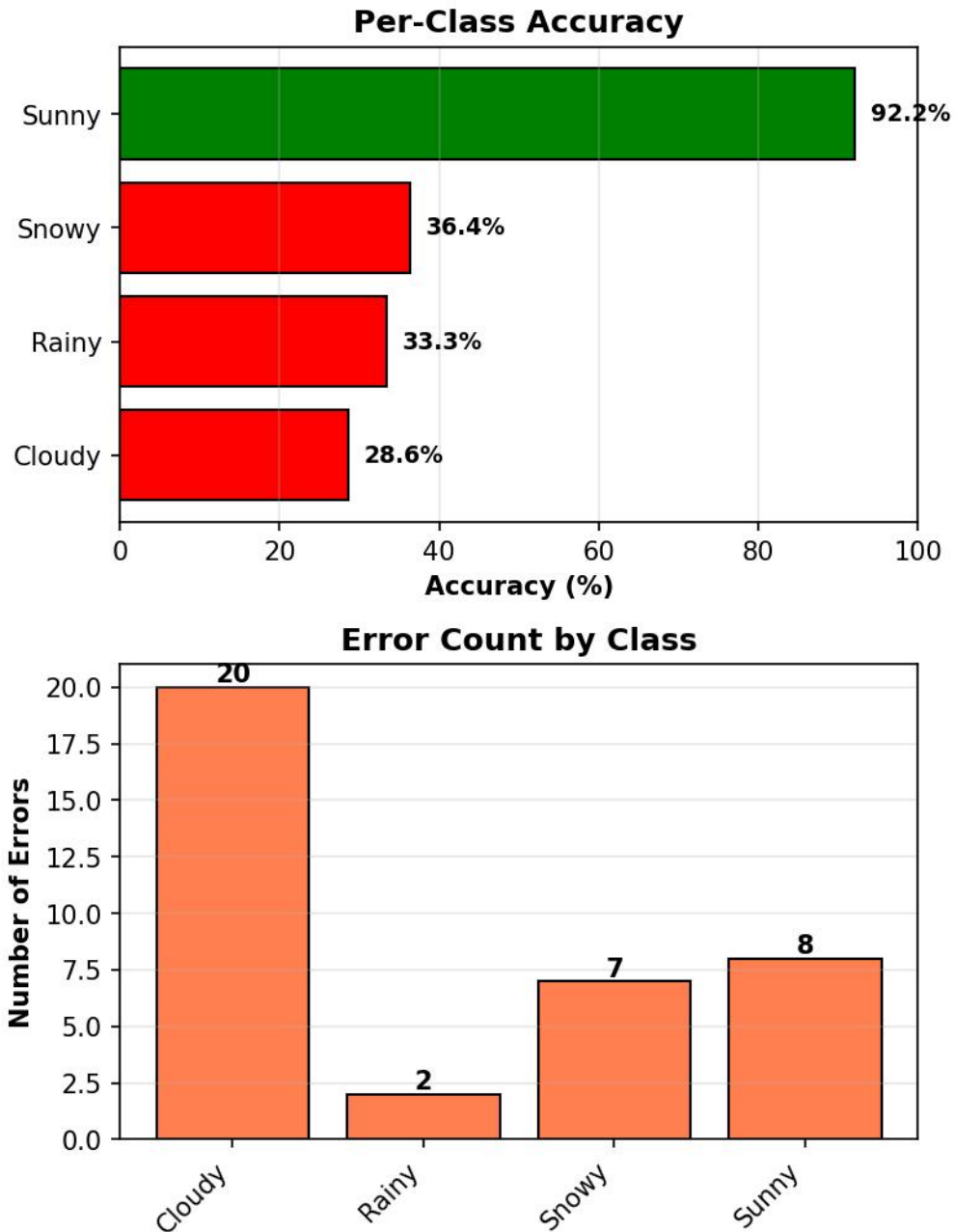


Figure 14: Per-class accuracy and error counts for the best model (CNN ResNet50).

3. Confusion Pattern Analysis (What mistakes are most common?)

We then examined the most frequent confusion patterns by counting how often each true label was predicted as another label. The most dominant misclassification was:

- **Cloudy → Sunny:** 19 errors (67.9% of Cloudy test samples)

Other noticeable confusions included:

- **Snowy → Sunny:** 4 errors (36.4% of Snowy)
- **Snowy → Cloudy:** 2 errors (18.2% of Snowy)
- **Sunny → Cloudy:** 5 errors (4.9% of Sunny)

This shows that many non-sunny weather conditions are being mapped into the Sunny class, especially Cloudy scenes.

4. Key Insights and Likely Causes (Data-driven)

Based on the per-class and confusion analysis, we identified several likely reasons behind these errors:

4.1 Class Imbalance Bias

The dataset is highly imbalanced in favor of Sunny images (Sunny dominates the dataset and the test set). As a result, the model learns stronger decision boundaries for Sunny and tends to predict Sunny when uncertain. This explains the frequent **Cloudy → Sunny** and **Snowy → Sunny** misclassifications.

4.2 Visual Similarity Between Classes

Several weather categories share overlapping visual features:

- **Cloudy vs Sunny:** Some cloudy images can still appear bright with patches of blue sky, leading the model to interpret them as Sunny.
- **Snowy vs Cloudy/Sunny:** Snow scenes may include bright lighting or washed-out skies, which can resemble Sunny lighting conditions.
- **Rainy vs Cloudy:** Both often include gray/white skies, making separation difficult without strong cues like raindrops, wet roads, or low visibility.

V. Conclusions and Discussion

Conclusions

This project studied weather prediction from images using a baseline model and two stronger models. The baseline KNN achieved 61.11% ($k=1$) and improved to 70.14% ($k=3$). The proposed models, SVM (best $C=100$) and CNN transfer learning (ResNet50), both achieved the best performance of 74.31%, improving the baseline by +4.17%. Overall, the results show that both feature-based machine learning and pretrained deep learning models can successfully classify weather conditions from images.

Limitations

The main limitations were the severe class imbalance (Sunny dominates the dataset), which caused high confusion such as Cloudy $\rightarrow$ Sunny, and the small number of minority-class samples (Rainy/Snowy). In addition, weather labels can be ambiguous, and using only accuracy may hide weak performance on rare classes.

Future Work

Future improvements include collecting a more balanced dataset, applying class weighting/oversampling, using stronger data augmentation, and fine-tuning the ResNet50 backbone. It would also be better to report macro-F1 and per-class recall instead of accuracy only.